A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 23

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 5 points below. **1** Harvest rate below the maximum sustainable yield – only economically valuable species (mahogany) selectively logged, not clear-felled **1** Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing – large seed trees and small trees left to grow **1** Buffer zones with no harvesting or machinery near rivers / wetlands **1** Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees **1** Each felled tree tagged and tracked to the retailer for FSC accreditation – rangers control illegal logging – tree nurseries replant over-exploited and ecologically important species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **maximum sustainable yield** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it **1** Employment as rangers, forestry and tourism workers – ecotourists use local restaurants, produce and souvenirs – offcuts given to local carvers **1** Benefits give the community an incentive to protect the forest – carbon sequestration payments from companies fund management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept – timber resource used over a longer period **1 Disadvantage:** more labour-intensive and expensive – more tracks needed per tree **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	

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Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm^2) **1** They use the polyp's CO2 and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently **1** They supply the polyp with glucose and oxygen - over 95% of its food energy – the polyp provides a protected habitat and nutrients **1** Polyps also trap plankton with cilia and stinging nematocysts – surplus CO2 is used to build the calcium carbonate skeleton **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** High light (shallow, clear water) for the algae – warm stable temperatures (about 18-30 °C) – low turbidity – constant salinity – well-oxygenated water **1** Found only where all these coincide (about 1% of ocean area) **1** Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **range of tolerance** → the range of conditions in which a species can survive	3
02.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Fisheries:** support over a quarter of marine species including fish, lobsters, conch **1 Erosion protection:** complex structure dissipates wave energy protecting low coral islands – coral sand builds shores **1 Medicinal discoveries:** AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts **1 Climate control:** carbonate skeletons sequester CO2 **1** Tourism (diving, snorkelling) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands – HEP schemes alter river levels flooding sandbank nests of turtles / birds – species with narrow tolerance (specialists) affected first – sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade **2** Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae **2** Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **abstraction** → water taken out of a river, lake or aquifer • **HEP** → electricity generated from falling water **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Reduces light penetration so submerged plants cannot photosynthesise **1** Blocks the gills of filter feeders such as bivalve molluscs – sediment smothers the bed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Rare species often have very specialised habitat requirements / narrow ranges of tolerance **1** A small change moves conditions outside their range while generalists survive – small populations cannot recover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - radio waves do not travel through seawater – acoustic or satellite tags transmit position	1

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Question	Marking guidance	Marks
04.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved **2** Radio / GPS tags or collars – leg rings / wing tags on birds – camera traps – DNA from droppings to confirm identity and diet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** The scatter / variability of results around each mean – about 68% of values lie within one SD of the mean **1** The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** t-test **1** Comparing two means of measured (continuous) data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2

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Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 30% probability that the difference occurred by random chance **1** Above the 0.05 (5%) level so the difference is not significant – the null hypothesis (no difference) is accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Two correlated variables may both be caused by a third variable that was not investigated **1** Significance only shows the result is unlikely to be chance – a mechanism / controlled experiment is needed for causation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **correlation** → a relationship in which the two variables change together • **causal** → one variable actually causing the change in the other **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
05.5	**1 mark each** 1 mark for each of the 2 points below. **1** Increase the number of replicates (reduces the effect of variability) **1** Standardise other variables (soil, water, light) more closely to reduce scatter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	

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Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Other MEDCs (USA, Canada, Australia) also had surpluses – Japan already had suppliers **1** Eastern Europe / USSR needed food but could not pay **1** Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** Quotas limiting production (eg milk) **1** Set-aside payments for taking land out of production **1** Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) **1** Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
06.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape **1** Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat **1** Beetle banks of tussocky grass - natural pest predators **1** Skylark nesting plots – wild bird seed / nectar plants - winter food and pollinators **1** In-field tree protection – controlled stocking density to reduce compaction – ditch management – dry-stone walls – woodland management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	

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Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Yield increases as inputs increase **1** But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Second unit:** 70 - 50 = 20 units **1 Fifth unit:** 94 - 90 = 4 units **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (units) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** The first units of input on each area give the largest yield increases **1** Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Some areas have much greater potential productivity so investment there is sensible **1** Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.5	**1 mark each** 1 mark for each of the 2 points below. **1** Yield per unit area is very high because of large inputs **1** But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	

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Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 3 points below. **1 Hard release:** individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) **1 Soft release:** gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds **1 Problems faced:** finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
08.2	**Any 5 of 7** 1 mark each for any 5 of the 7 points below, in any order, maximum 5. **1** Large enough suitable habitat for a viable population **1** Reliable food supply **1** Low predation risk / original threat removed **1** Suitable breeding sites **1** Water **1** Support of the local human population **1** Official / legal protection of habitat and species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	5
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed **1** Identifies causes of failure so future releases can be improved / extra support given **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	

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Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** The processes (eg regulation of atmospheric gases by photosynthesis and respiration) act in a dynamic equilibrium and cancel each other out **1** Because there is no overall change their actions go unnoticed / their importance is not appreciated until they are lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Invertebrates, fungi and bacteria break down / decompose dead organic matter, releasing nutrients for plant growth **1** The organic matter and humus produced hold the soil together (reducing erosion) and regulate the water cycle, giving more even river flow and less flooding **1** Organic acids from decomposition aid the weathering of rock, producing more soil and releasing more nutrients **1** Microbes carry out the processes of the carbon, nitrogen and phosphorus cycles so wastes do not build up and nutrients are not depleted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Trees reduce wind velocity so wind pollination would be ineffective / unreliable over long distances **1** Insects search over long distances for flowers so dispersed populations can be pollinated, and less pollen needs to be produced (energy saving) – specialised flowers reduce pollen being wasted on other species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.4	**1 mark each** 1 mark for each of the 3 points below. **1 Dispersal is not haphazard:** animals live in the same habitat as the plant so are likely to carry seeds to locations where they can survive **1** Seeds can be carried longer distances than by wind and can be larger, containing more nutrients for the seedling **1** Seeds passing through the gut are deposited in faecal material which acts as a fertiliser, and passage through the intestine may stimulate germination **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	

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Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Water:** catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already – limits in arid regions • **Minerals:** exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging – limits: dispersal, alloys, energy of collection • **Waste management:** repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste – EU targets on packaging, batteries, WEEE • **Pollution control:** shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours) – limits for legacy pollution • **Energy:** renewables, storage, energy density technologies, low-temperature manufacturing, CHP – limits of intermittency, transport fuels • **Food:** natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture – tension with maximum yield for a growing population • **Land uses supporting ecosystems:** urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • **Judgement:** high applicability in some sectors (water, waste), partial in others (minerals, energy for transport) – needs policy support **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	
	Taking water, waste and pollution control: water supply is already close to circular, because the hydrological cycle returns what is used. Catchment management, conservation, metering, grey-water re-use, re-use of treated effluent, aquifer recharge and sustainable urban drainage close the loop further, with the main limit in arid regions where abstraction exceeds recharge. Waste management can apply the principles almost completely: repair, re-use and refurbishment extend product life, recycling returns materials, composting and anaerobic digestion return nutrients and yield biogas, residual waste supplies energy and landfill is minimised, driven by EU targets on packaging, batteries and WEEE; the limits are mixed materials that cannot be separated and the energy cost of collection.	
	Pollution control changes character under circular principles: instead of treating wastes after production - flue-gas desulfurisation, dispersants - the aim becomes non-release, through fuel desulfurisation, fuel cells and electric vehicles, and industrial symbiosis, as at Kalundborg where one plant's waste heat, gypsum, fly ash and yeast become neighbours' raw materials. Land use can support ecosystems with green infrastructure, gardens, permeable surfaces and habitat in redeveloped docks.	
	Applicability is high for water and waste because natural cycles already exist to copy; it is partial for pollution control because many processes still depend on combustion, and it needs policy support - targets, producer responsibility, planning - to overcome the economics of linear systems. The principles can be applied widely, but not evenly.	
	Weaker answer - Level 1 (3 marks)	
	The circular economy means reusing and recycling things instead of throwing them away. For water this means saving water and reusing grey water. For waste management it means recycling more and putting less in landfill. For pollution control it means not producing pollution in the first place. The circular economy can be applied to all of these to make them more sustainable.	
	What makes the difference	
	• Level 1 gives one idea per area; Level 3 gives several specific measures for each and names the limit in each case.	
	• Level 3 uses a case study (Kalundborg) and policy examples (EU targets) to show application, not just principle.	

Do not write outside the box

Question	Marking guidance	Marks
	• 'Evaluate the extent' - Level 3 compares applicability across the three areas and explains the differences; Level 1 says it applies to all.	
10.2	1 mark for the point below. **1** Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiosis** → a close relationship in which both species benefit	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Sustainability definitions – population and per capita impact – attitudes of rural and urban societies • **Negative feedback:** cloud cover and albedo, CO_2 and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • **Tipping points:** rare under negative feedback – large events (Permian methane hydrate) – self-sustaining change from human activities • **Diversity and resistance:** reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides) – links to climate change chapter – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Importance:** resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • **Causes:** agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY – trends by region • **Consequences:** loss of resources, biodiversity and species, fragmentation – hydrological changes – soil impacts – climate impacts (albedo, carbon, rainfall) • **Productivity vs biodiversity:** non-indigenous species, monocultures, close planting, simple age structure, clear-felling • **Sustainable management:** MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China) – Rio Bravo case study	25
	• **Evaluation:** economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	

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